

NovaRetail – AI Assistant Architecture Review

Date: January 21, 2026

Main Topics

1. Embedding Model Evaluation

Emre compared: all-MiniLM-L6-v2 bge-small-en OpenAI embeddings Findings: OpenAI embeddings produced the best retrieval quality Local models reduced operational cost significantly Burak requested cost projections for 1000 daily active users.

2. Chunking Strategy Problems

The team identified poor retrieval performance in long meeting transcripts.

Observed issues: important decisions split across chunks duplicated retrieval results missing contextual continuity

Suggestions: hierarchical chunking metadata tagging overlap increase from 100 → 250 characters No final decision reached.

3. Agent Workflow Proposal

Selin proposed adding: action item extraction meeting summarization risk identification workflows Zeynep warned against enabling autonomous email generation before legal review.

Open Questions

- Should meeting recordings also be indexed?
- How should deleted documents be removed from vector storage?
- Is document versioning required for audit compliance?

Action Items

- Emre: Test hierarchical chunking
- Can: Investigate vector DB cleanup strategies
- Zeynep: Review legal implications of AI-generated summaries